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NATIONAL LEVEL HACKATHON



TEAM NAME: **M&Ms**



PROBLEM STATEMENT:

Students struggle to find accurate answers from their own study materials.
Existing AI tools use external knowledge, leading to irrelevant or incorrect responses.



TEAM MEMBERS: Mukul Yadav (Leader) | Mayank Chaudhary | Medha Pandey | Mayank Sharma | Mayank Kumar



COLLEGE: **ABES Engineering College**

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PROBLEM STATEMENT



Students struggle to organize scattered study materials and quickly retrieve relevant information from multiple PDFs. Existing AI tools rely on external knowledge, leading to **inaccurate or irrelevant responses**.



THE RESULT

Students lose **valuable time, confidence, and productivity** because academic knowledge remains fragmented and difficult to access.

SOLUTION

We built **SARASWATI**

An AI-powered academic workspace that provides accurate, context-aware answers using **only your uploaded documents**.



Upload

Upload your academic PDFs and notes securely.



Organize

Organize them into subject-wise folders for better structure.



Ask

Ask any question from your documents in natural language.



Get Accurate Answers

AI answers using only your uploaded content—no external knowledge.



Our AI never uses external knowledge.



Accuracy



Trust



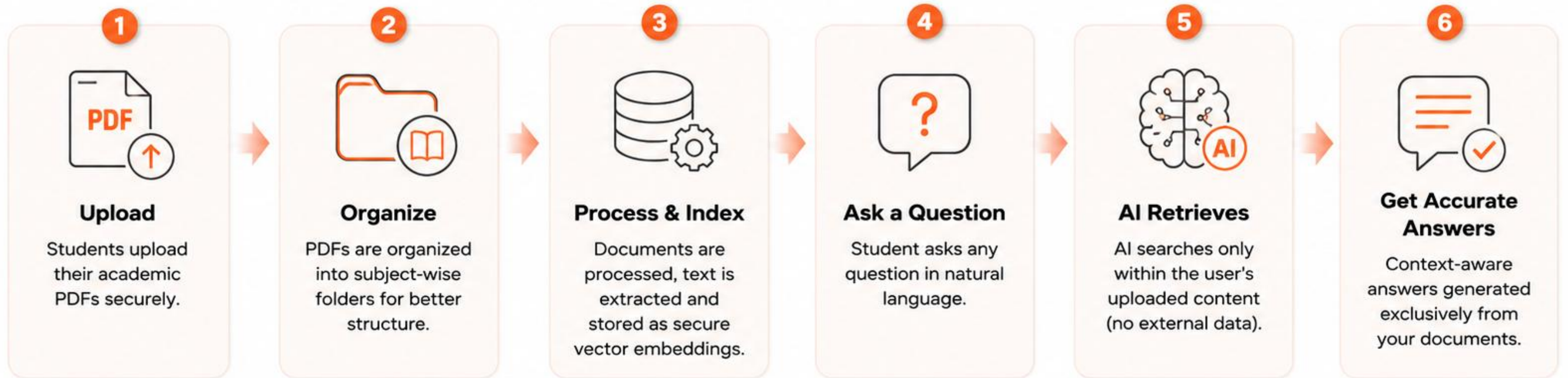
Privacy



Context Awareness

FLOW OF SOLUTION

How **SARASWATI** works – From your documents to accurate answers.



Our AI Never Uses External Knowledge

Every response is generated only from your uploaded academic documents, ensuring accuracy, trust, privacy and complete context awareness.



Accuracy



Trust



Privacy



Context Awareness

TECH STACK

FRONTEND



React

Building a fast, responsive and interactive user experience.



Lucide React (Icons)

Beautiful & consistent icons for a modern UI.

BACKEND



FastAPI

High-performance Python framework for building robust APIs.



LangChain

Framework for building LLM applications with chains, memory and tools.



Uvicorn

Lightning-fast ASGI server to run FastAPI.

AI & RAG



Mistral (LLM)

Powerful and efficient LLM for intelligent, accurate responses.



PyMuPDF (PDF Parsing)

Fast and reliable PDF parsing and text extraction.



Recursive Character Text Splitter

Smart chunking for better context understanding.



ChromaDB (Vector Database)

High-performance vector database for similarity search.



Why this stack?



High Performance



Scalable & Reliable



Developer Friendly



Built for Innovation

UNIQUE SELLING PROPOSITION (USP)

Why **SARASWATI** is different and better.

KEY FEATURES & INNOVATION



Secure PDF Upload

Upload and store your academic PDFs securely in one place.



Subject-wise Organization

Organize documents into subject-wise folders for better structure.



AI Chat with Your Docs

Ask any question and get answers strictly from your uploaded documents.



Context-Aware AI

Advanced retrieval ensures precise, relevant and concept-specific responses.



Privacy First

Your data stays yours. No sharing, no training on your documents.



Fast & Reliable

Quick search and accurate answers to save time and boost productivity.

SARASWATI does not use any external knowledge.

Every answer comes **only** from your uploaded documents.



100% Document-Grounded Responses

AI answers are generated strictly from your uploaded PDFs—no external data, no hallucinations.



Built for Students, by Students

Designed specifically for academic needs—notes, textbooks, assignments, and exam preparation.



Trust, Accuracy & Reliability

Delivering accurate, reliable and referenceable answers you can trust.



Your Knowledge. Your Space.

A personalized AI workspace that understands only your content and context.

FEASIBILITY & COMPETITORS



FEASIBILITY

Technically achievable with current technologies and real academic demand.



Technically Feasible

Built using proven AI/ML models, vector databases, and secure cloud infrastructure.



High Demand

Students across engineering and universities constantly struggle with document management.



Data Security

All documents are stored securely; AI accesses only user-specific data.



Scalable Solution

Designed to scale with more users, subjects, and document types seamlessly.



COMPETITORS COMPARISON

SARASWATI stands out by ensuring accuracy, privacy and context-aware responses.

FEATURES	SARASWATI (Our Solution)	ChatGPT (GPT-4)	Google Gemini	Microsoft Copilot
 Uses only your uploaded documents				
 100% Accurate & Context-Aware		 (Sometimes)	 (Sometimes)	 (Sometimes)
 Privacy & Data Security		 (Limited)	 (Limited)	 (Limited)
 No External Knowledge Used				
 Built for Academic Use Cases		 (General)	 (General)	 (General)

RESEARCH & REFERENCE

Building **SARASWATI** on proven research and cutting-edge technologies.

OUR RESEARCH APPROACH



1. Document Ingestion

PDFs are uploaded and parsed using PyMuPDF to extract clean text.



2. Text Chunking

Text is split into meaningful chunks using Recursive Character Text Splitter.



3. Retrieval (RAG)

Relevant chunks are retrieved from ChromaDB based on the user query.



4. Generation

Mistral (LLM) generates accurate answers strictly from the retrieved context.



Our system follows a strict **RAG** pipeline to ensure that every answer is grounded in the student's own documents – **no external knowledge** used.

REFERENCES & TECHNOLOGIES USED



RAG (Retrieval-Augmented Generation)

Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.



Vector Similarity Search

Pinecone, Weaviate, FAISS – Scalable similarity search techniques.



Embedding & Semantic Search

Reimers, N. & Gurevych, I. (2019). Sentence Transformers for Semantic Search.



PDF Parsing

PyMuPDF Documentation – Fast and reliable PDF text extraction.



Vector Database

ChromaDB – Lightweight and efficient vector database.



LLM (Mistral)

Mistral AI – Efficient, high-quality language model for generation.



Best Practices

OWASP Top 10, Data privacy principles and secure AI development practices.



GIT REPOSITORY

Explore our source code, documentation and latest updates.



GitHub Link:

<https://github.com/mukul2007yadav-lab/Saraswati>